

No. of Functional Features Included in the Solution

Date:	08 October 2026	Team ID:	UNM1611JASON
Project Name:	AI BlogNest API	Maximum Marks:	5 Marks

Functional Features Overview:

This document inventories all 8 functional features developed and verified in the AI BlogNest API solution, detailing their operational purpose, module location, completion status, and mark contributions.

S.No	Feature Name	Feature Description	Module / File	Status	Marks
1	User Registration	Registers new account with bcrypt password encryption (10 salt rounds) and duplicate email prevention.	authController.js	Done	0.6
2	User Login & Token	Authenticates email and password, returning signed HMAC-SHA256 JWT access token valid for 7 days.	authController.js	Done	0.6
3	User Profile Lookup	Decodes Bearer token via authMiddleware and retrieves current authenticated user profile.	authController.js	Done	0.6
4	Blog Post Creation	Validates title, content, category; links blog document to req.user._id and authorName.	blogController.js	Done	0.6
5	Blog Query & Populate	Retrieves all blogs sorted descending by date with populated author details, or single blog by ID.	blogController.js	Done	0.6
6	Protected Update & Delete	Restricts blog mutations strictly to the original author with 403 Forbidden check.	blogController.js	Done	0.6
7	Gemini AI Blog Generator	Generates 100-150 word post via Gemini 1.5 Flash, cleans markdown, auto-categorizes, and saves.	aiController.js	Done	0.7
8	Gemini AI Summarizer	Accepts long article body and returns single crisp executive summary paragraph.	aiController.js	Done	0.7

Metric	Count / Value	Metric	Count / Value
Total Features Planned	8	Additional / Nice-to-Have Features	2
Total Features Implemented	8 (100%)	Features Tested & Verified	8 (100%)
Core / Must-Have Features	6	Overall Functional Marks	5.0 / 5.0

No. of Functional Features — Category Breakdown & Endpoints

Feature Category Breakdown:

S.No	Category	Features in Category	Concrete Implementation in AI BlogNest API
1	API Testing Client (UI)	Thunder Client Suite	Interactive Postman collection v2.1 with configured environment variables {{base_url}}, {{token}}, {{blogId}} for rapid UI verification.
2	Backend / Business Logic	3 Features	Express routing engine, dynamic keyword category classifier, regex text sanitizer, and centralized error logging middleware.
3	Database / Storage	2 Features	Mongoose User & Blog schemas with indexed email fields, automatic timestamps, and relational author population.
4	AI / LLM Integration	2 Features	Google Generative AI SDK integration dispatching custom prompt templates to gemini-1.5-flash for blog generation and text summarization.
5	Security & Auth	2 Features	Bcrypt salted password encryption, JWT Bearer token generation, and author authorization guards protecting blog modification.

Endpoint Mapping Reference:

HTTP Method & Path	Auth Required	Input Payload	Output Response
POST /api/auth/register	No	name, email, password	201 Created + user {id, name, email}
POST /api/auth/login	No	email, password	200 OK + { token }
GET /api/auth/profile	Yes (Bearer)	None (Header token)	200 OK + user profile object
POST /api/blogs	Yes (Bearer)	title, content, category	201 Created + blog document
GET /api/blogs	No	None	200 OK + array of blogs with author info
PUT /api/blogs/:id	Yes (Author)	title?, content?, category?	200 OK + updated blog object
DELETE /api/blogs/:id	Yes (Author)	None	200 OK + { message: 'Blog deleted' }
POST /api/ai/generate-blog	Yes (Bearer)	topic, category?	201 Created + Gemini generated blog
POST /api/ai/summarize	No	content	200 OK + { summary: text }